

But when I told him what had happened, Bezos looked horrified. He did not say “I’m so sorry.” He did not say “Do you need anything?” Instead, he made a face, and in an instant, an aide came and whisked him away. When presented with the opportunity for empathy, even performative empathy, he chose escape.

A few hours later, on the private plane home, a famous movie producer offered my wife a blanket. My children’s faces were covered in spots. Under my fingernails, red welts were beginning to rise.

The world has always been run by rich men. The robber barons of the Gilded Age were known for their ruthlessness in the accumulation of wealth—hiring Pinkertons to shoot striking unionists. But they directly engaged with the world around them, using their wealth and power to muscle it into its most profitable form. And although today’s billionaires are clearly manipulating society to maximize their own profit, something else is also happening—a disassociation from the reality of cause and effect, from meaning and history. These men no longer feel the need to change the world in order to succeed, because their success is guaranteed, no matter what happens to the rest of us.

“I’m finished,” yells Daniel Plainview, perched happily on the polished floor of his own celestial kingdom. Though he has just committed a crime, he has never felt so free.

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Lessons learned from 1 year of Judge-LLM Development

By

[Zeno Belligoli](#)

,

[Antonio Castelli](#)

,

[George Chouliaras](#)



The increasing adoption of Large Language Models (LLMs) across various industries has made evaluating LLM-powered applications (also called Generative AI applications) a critical necessity.

LLMs are pre-trained on a vast amount of text and can therefore be used to perform a wide range of tasks based on a natural language input called **prompt**. These tasks range from extracting information, answering complex questions, summarizing documents, generating creative content, translating languages, writing code etc.

with the air of a man who believed that nothing matters—poverty, chaos, [human suffering](#). He was having fun. It didn’t even matter that the [entire destructive exercise](#) ultimately yielded no practical financial gains. For him, the outcome was a foregone conclusion: He could only win, because losing had lost its meaning.

Since the 2024 election, there has been a philosophical shift on the right, and especially among tech billionaires, to vilify the idea of empathy. Musk has called empathy “the fundamental weakness of Western civilization.” He sees it as a weapon wielded by liberal society to bludgeon otherwise rational people into operating against their own interests. Empathy is something done to you by others—a vulnerability they exploit, a back door through which they gain access to your resources and will. This [rejection of empathy](#) as a human value gives cover to people who don’t want to feel anything at all. If empathy is the problem, then lack of it isn’t a deficiency—it’s [an advantage](#).

[Elizabeth Bruenig: The conservative attack on empathy](#)

I finally met Bezos on the last day of Campfire, at lunch, after my wife had broken her wrist. I went over to thank him for having us, and he asked how our Campfire experience had been. I told him that it was great, but that unfortunately my wife had broken her wrist that morning when she slipped on the wet grass while kicking a ball with our 6-year-old son.

The night before, we’d all stood by the pool at the beach club watching a cadre of synchronized swimmers execute a flawless water routine. I had spoken with a famous novelist, who said, “I just don’t understand why I’m here.” A famous rock star was about to start an acoustic set. The famous chef had made paella. Somewhere deep under my skin, a brutal pox was beginning to form.

The next morning, my wife fell, and I found myself in a black SUV with a team of private-security contractors, who whisked us to the back entrance of a Santa Barbara emergency room, where she was seen and treated right away. We made it back in time to watch the Supreme Court justice Zoom in from Washington, D.C.

How was your Campfire? Bezos asked me an hour later, and because I am an honest person, and because I have been a host myself, I decided he would want to know that there had been a problem, but that his team had reacted quickly and been extremely helpful. To be clear, I was in no way blaming him, nor was I shaking down the richest man on Earth. Instead, I was simply offering Bezos, also a husband and father, a brief human connection.

that his financial and social value could be affected by negative publicity. He still believed that his actions had consequences. He had not yet freed himself—the way Daniel Plainview freed himself—from the rules of men.

Eight years later, Bezos and two of the world’s other richest men—Mark Zuckerberg and Elon Musk—have clearly left the world of consequences behind. They float in a sensory-deprivation tank the size of the planet, in which their actions are only ever judged by themselves.

The closer I’ve gotten to the world of wealth, the more I understand that being truly rich doesn’t mean amassing enough money to afford superyachts, private jets, or a million acres of land. It means that everything becomes effectively free. Any asset can be acquired but nothing can ever be lost, because for soon-to-be trillionaires, no level of loss could significantly change their global standing or personal power. For them, the word *failure* has ceased to mean anything.

Adam Serwer: How America chose not to hold the powerful to account

This sense of invulnerability has deep psychological ramifications. If everything is free and nothing matters, then the world and other people exist only to be acted upon, if they are acknowledged at all. This is different from classic narcissism, in which a grandiose but fragile self-image can mask deep insecurity. What I’m talking about is a self-definition in which the individual grows to the size of the universe, and the universe vanishes. Asked recently if there is any check on his power, President Trump—himself a billionaire, and by far the richest president in American history—said, “Yeah, there is one thing. My own morality. My own mind. It’s the only thing that can stop me.” Not domestic or international law, not the will of the voters, not God or the centuries-old morality of civic and religious life.

Decades of research in developmental psychology have shown that moral reasoning develops through consequences—not punishment, necessarily, but experiencing the effects of your actions on others, receiving honest feedback, having to accommodate reality as it actually is rather than as you wish it to be. It’s not that the wealthy become evil; it’s that their environment stops teaching them the things that nonwealthy people are forced to learn simply by living in a world that pushes back. When you can buy your way out of any mistake, when you can fire anyone who disagrees with you, when your social circle consists entirely of people who need something from you, the basic mechanism by which humans learn that other people are real goes dark.

Thomas Chatterton Williams: The very powerful men who think introspection is dumb

When Peter Thiel said, “I no longer believe that freedom and democracy are compatible,” he wasn’t talking about your freedom. He was talking about his own. You don’t exist. When Musk took a chainsaw to the federal government as part of the inside joke he called DOGE, he did so

However, unlike traditional Machine Learning (ML) models, the nature of LLMs brings inherent challenges that must be carefully considered:

- LLMs can **hallucinate**, i.e. can generate outputs which are factually incorrect or nonsensical while presenting them with high confidence
- LLMs can fail to **follow the instructions** specified in the prompt no matter how detailed they sound to a human reader
- Usage of LLMs has a **non-negligible cost** whether the model is open-source and served in-house or closed-source and accessed via a paid API.

To maximize the potential of Generative AI (GenAI) applications and mitigate the risks associated with them, we built a framework capable of thoroughly evaluating the performance of an LLM on a specific task in a nearly automated way. This framework is based on the concept of **LLM-as-judge**, i.e. on the usage of a more powerful LLM to evaluate the “target” LLM.

The LLM-as-a-judge framework

The main difference between the evaluation of traditional ML models and LLMs is that in most of the cases we are dealing with generative tasks for which either no single ground truth exists or is hard / expensive to obtain. When we ask an LLM to write a description of a property, or to summarize a long text, we don’t really have an unambiguous reference that we can use for comparison. Moreover we can’t get this at scale for different topics or subjects.

Therefore the measurement of text attributes like clarity, readability, toxicity or factual accuracy becomes more challenging. It would be possible, in theory, to employ human experts to review every generation and provide an estimation of the metric under study. However, this process would be time consuming and expensive to be **practically infeasible**.

The LLM-as-judge approach, requires human involvement **only once** (unless the production distribution changes), to carefully annotate the so-called **golden dataset** which must be large enough to be representative of the data distribution in production (the best practices for golden dataset creation are described in the next section).

Once we have the golden dataset with labels, we can prompt (or fine-tune) an LLM to replicate human judgement as much as possible. Once this is achieved, (e.g. when the agreement between the judge LLM and the human annotation has an accuracy above a certain threshold) the judge LLM can be employed to score the predictions of a second LLM (*target LLM*) on other datasets.

This allows the continuous monitoring of the performance of a GenAI application in production in a scalable way with minimal human involvement.

Since the judge-LLM solves a classification task (e.g. is the text clear or not? is the text factually accurate or not?) getting golden labels is much easier. Hence, we can evaluate its performance using standard metrics like accuracy, f1-score, precision or recall, to monitor the performance of the target LLM. The process of judge-LLM development and usage is depicted in Figure 1.

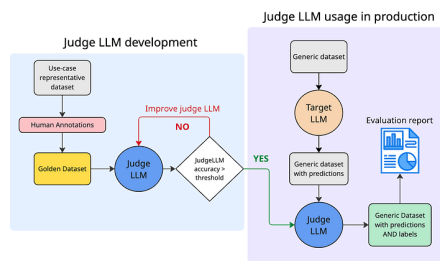


Figure 1: Judge-LLM development and deployment cycle. The left box (light-blue) shows the judge-LLM development phase. This phase always starts with a golden dataset with labels provided by a fully trusted source (e.g. human experts). A base LLM model is prompted or fine-tuned to imitate human label scoring. When the judge-LLM reaches a certain accuracy with respect to the golden labels, it can be deployed in production (right purple box) to score any other similar dataset, for example to evaluate the output of a target LLM. An evaluation report can be produced using the predictions of the target LLM and the label provided by the Judge LLM.

Golden Dataset Creation

The main aim of a golden dataset is to **accurately assess the ability** of a judge-LLM, to evaluate a production LLM in a particular evaluation metric. Due to this, a **high quality** golden dataset should meet the following criteria:

- It should represent the *production distribution*, since we want the Judge LLM evaluation to be as close to production as possible.
- It should include *golden* labels, e.g. labels with high quality. This can be proxied by high inter-annotator agreement among the annotators of the labels.

That’s how the weekend started. Here’s how it ended: My wife broke her wrist slipping on wet grass, and both children and I came down with hand, foot, and mouth disease. This is not a joke. One of us went home with her arm in a sling; the other three developed itchy, painful red blisters all over our faces and extremities. If you’re looking for a sign from God as to whether hanging out with the richest man on Earth is right for you, pay attention when he sends you not one plague, but two. Suffice it to say we have never been back to Campfire, nor have we ever been invited.

At drinks on the second night, the head of a major talent agency asked me what I thought of the weekend. I said, “I’ve spent my whole career trying to figure out how the world works. I didn’t realize I could just come here and ask the people who ran it.” On some level I was kidding. The lead singer of an alt-country band didn’t run the world, nor did a noted author who would later be accused of impropriety. But finding myself at that resort by exclusive invitation, I now knew exactly what people meant when they talked about the elite.

Sitting in the lecture hall, pencils out, listening to a famous chef explain his humanitarian work, it was easy to feel like the solution to the world’s problems lay within our grasp. And yet, looking around at faces I had only ever seen in a magazine or on-screen, I had an unsettling revelation: This is the hubris of accomplishment. To be declared a genius at one thing is to begin to believe you are a genius at everything.

Here we were, 80 individuals with a combined net worth that was greater than a small city’s yet infinitesimal compared with the wealth and dominion of our host. How did he view this exercise—as a first step toward changing the world, or as a performative display of his reach and influence?

Bezos was everywhere that weekend—in a tight T-shirt, laughing too loudly, arms thrown around his teenage sons. He had recently become the world’s second centibillionaire, his net worth hovering somewhere around \$112 billion, about half of what it is today. That number, previously unimaginable, had made him unique on a planet of 8 billion people, and you could feel it in the room. Even the richest and most famous among us were drawn to the energy of this impossible wealth.

Martin Baron: Where Jeff Bezos went wrong with *The Washington Post*

Though we didn’t know it at the time, Bezos’s first marriage would be over a few weeks later. My defining impression of his wife that weekend was sadness, even though Bezos made a big show of performing the role of family man. In hindsight, it is that performance that sticks with me. The Jeff Bezos of 2018 acted as if he still believed that people’s impression of him mattered,

In 2018, I was a guest at Jeff Bezos’s Campfire retreat in Santa Barbara, California. It’s an annual event in which the Amazon founder invites 80-plus guests—celebrities, artists, intellectuals, and anyone else he thinks is interesting—to spend three nights at a private resort. I had recently been approached by Amazon about moving my film-and-television business over from Disney, and although I had declined (or maybe *because* I had declined), Bezos’s team invited me to Campfire, perhaps keen to impress me with the power of his reach.

From the March 2024 issue: The rise of techno-authoritarianism

On a warm October Thursday, a fleet of private jets was dispatched to airports in Van Nuys and New York to shepherd guests to Santa Barbara in style. At that point I had only a vague sense of who else was coming—famous people, rich people, influential people, and me. A guest list, I was told, would be given to us once we arrived. Families were invited; an on-site nanny would be provided *for each child*.

So my wife and I got our two children from Austin to Los Angeles and took a 45-minute jet ride north, with a television mogul and a comedian on board. Bezos had bought out the entire Biltmore resort for the weekend, as well as the beach club across the street. He had brought in a security firm from Las Vegas to ensure our safety and privacy. Even the weather felt expensive, and when we were shown to our rooms, the designer gift bags we found were filled with luxury goods.

Alexandra Petri: The 10 things the Bezoses are almost certainly grateful for each morning

Each morning, we gathered in a lecture hall to hear presentations. If you’ve ever seen a TED Talk, you understand the format. The year I went, a sitting Supreme Court justice was interviewed, and a neurologist talked about technological advances in prosthetics. In the afternoons and evenings, we were encouraged to exchange ideas over drinks and four-course meals, with no set purpose—to network, in other words, with some of the most rarefied talent on Earth. The most common question I heard was “Why am I here?”

“Why am I here?” asked the 1980s hair-metal singer. “Why am I here?” asked the Pulitzer Prize–winning novelist, the famous anthropologist, the presidential historian. Only the movie stars and the billionaires didn’t ask: They had done this kind of thing before. It turns out there is a circuit of idea festivals. Many tech billionaires host one, and if you find yourself on the right list, you can spend much of the year traveling the world, eating Wagyu, and discussing how to make the world a better place with the most famous talk-show host in history.

Low inter-annotator agreement usually indicates ambiguity in the metric definition and should be resolved by calibration among the annotators and by adding clarifications in the definition until the agreement is acceptable.

Since text attributes often do not have an objective and unique true value, achieving high-quality annotations is not a trivial task. To make fair comparisons between different versions of a GenAI application it is of utmost importance to have a **standardized annotation protocol** which is rigorously followed. Below we provide both a **basic** and an **advanced** annotation protocol for creating the golden dataset.

Basic Annotation Protocol

The basic annotation protocol can be used when only a single annotator is available.

1. Metric Definition

Define the metric definition with the business owner in the most unambiguous way possible.

- Write a **clear** definition of the metric as objective as possible. Prefer **binary** metrics or categorical metrics with a few classes, since LLMs struggle with continuous scores. If you’re forced to use a continuous metric, consider binning into a few coarse ordinal values e.g. ‘high’, ‘medium’, ‘low’.
- Write **annotation instructions**, using the metric definition. These are the instructions a human should follow in order to provide the ground truth label. Describe the task **clearly** and **objectively**. Avoid vague words (such as major, minor, a few). Provide scored examples and include edge cases. Specify what tools the annotators can and cannot use (Search engines, ChatGPT, etc).

2. Pilot Annotation

Collect a few examples (~50) and annotate them according to the aforementioned metric definition. The annotators should carefully follow the instructions provided, without bringing any previous knowledge or bias on the task. The annotators at this stage can be selected from business owners or from other technical people involved. The aim of this step is to ensure that we have a clear metric definition and that the annotations are aligned with it.

Once the annotations are done, it is recommended that they are **reviewed** by a pool of domain experts to assess if they align with the guidelines. If issues or disagreements are found, they should be resolved, the metric definition should be updated and the pilot annotation should be repeated until full consensus is reached.

3. Full Annotation by a single annotator

Once the annotation of the first small sample during the pilot annotation is finalized and reviewed, the examples can be shared to the annotator as guidance in order to annotate a full dataset. The full dataset should consist of **ideally 500- 1000 examples** at least. This is because part of the dataset will be used for tuning the judge-LLM (validation set) and another part as a holdout set in order to evaluate its performance.

4. Quality Review

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A subject matter expert reviews a **representative** sample (~10%) of the full annotation. If a significant percentage (> 10%) of the labels in the sample are incorrect, annotation should be repeated. The quality thresholds can be adjusted according to the organization's needs and should be communicated and agreed upon with the annotators before the full annotation.

Advanced Annotation Protocol

The advanced annotation protocol requires more effort but also guarantees higher quality annotations than the basic one.

1. Metric Definition

Same as in the basic protocol.

2. Pilot Annotation

Same as in the basic protocol.

3. Full Annotation by multiple annotators

Instead of having a single human annotator providing the label for a given example, we have multiple human annotators (3+) providing a label for each example. Depending on the use case, an aggregation function must be defined to obtain the final label. The aggregation function should take an array of labels and return a single one. In principle any

Navigating the dramatic changes of the next few decades while trying to avoid world wars and murderous ideologies is difficult enough without muddying our thinking.

What I Learned About Billionaires at Jeff Bezos's Private Retreat

NOAH HAWLEY · 02 MAY 2026 · [SOURCE](#)

For the richest men on Earth, everything is free and nothing matters.



(Illustration by Tim Enthoven)

At the end of Paul Thomas Anderson's 2007 movie, *There Will Be Blood*, Daniel Day-Lewis's oil-baron character, old now and richer than Croesus, beats Paul Dano's preacher to death with a bowling pin. Dano's Eli Sunday, a nemesis of Day-Lewis's Daniel Plainview during his seminal, wealth-building years, has come to sell Plainview the oil-rich land that he once coveted. But Plainview doesn't need the land anymore, because—as he explains in one of the most famous monologues in modern cinema—he has sucked out all the oil hidden beneath it from an adjoining property, like a milkshake.

Desperate for money, Eli begs for a loan. Instead, Plainview chases him around a bowling alley and murders him with great enthusiasm. Once it's over, a butler comes to see what all the noise was about. "I'm finished," Plainview yells.

No matter how many times I watch that movie, and I watch it a lot, I have never once taken those words to mean *I'm done for. There will now be consequences for my actions*. Quite the opposite: They mean that Plainview has completed his journey, through the acquisition of wealth and power, to a realm outside the moral universe. He's finished, in other words, pretending that the rules of human society apply to him.

Why many AI luminaries tend to anthropomorphize

Perhaps I am fighting windmills, or rather a self-selection bias: A fair number of current AI luminaries have self-selected by their belief that they might be the ones getting to AGI - "creating a god" so to speak, the creation of something like life, as good as or better than humans. You are more likely to choose this career path if you believe that it is feasible, and that current approaches might get you there. Possibly I am asking people to "please let go of the belief that you based your life around" when I am asking for an end to anthropomorphization of LLMs, which won't fly.

Why I think human consciousness isn't comparable to an LLM

The following is uncomfortably philosophical, but: In my worldview, humans are dramatically different things than a function . For hundreds of millions of years, nature generated new versions, and only a small number of these versions survived. Human thought is a poorly-understood process, involving enormously many neurons, extremely high-bandwidth input, an extremely complicated cocktail of hormones, constant monitoring of energy levels, and millions of years of harsh selection pressure.

We understand essentially nothing about it. In contrast to an LLM, given a human and a sequence of words, I cannot begin putting a probability on "will this human generate this sequence".

To repeat myself: To me, considering that any human concept such as ethics, will to survive, or fear, apply to an LLM appears similarly strange as if we were discussing the feelings of a numerical meteorology simulation.

The real issues

The function class represented by modern LLMs are very useful. Even if we never get anywhere close to AGI and just deploy the current state of technology everywhere where it might be useful, we will get a dramatically different world. LLMs might end up being similarly impactful as electrification.

My grandfather lived from 1904 to 1981, a period which encompassed moving from gas lamps to electric, the replacement of horse carriages by cars, nuclear power, transistors, all the way to computers. It also spanned two world wars, the rise of Communism and Stalinism, almost the entire lifetime of the USSR and GDR etc. The world on his birth looked nothing like the world when he died.

aggregation function can be used. If labels can be converted into integers and ordered any function like min, max, or average can be used. If labels have no orders (e.g. geographical location) majority vote can be used.

Examples where the annotators **disagree** can be treated in multiple ways, depending on what is more appropriate for the specific problem.

a) Add a “not sure” class: map to the “not sure” class all examples without 100% consensus

b) Add a weight < 1: For all examples without 100% consensus select the majority class among the annotated ones (a random one if there is no majority) and add a weight to the example equal to $n_{\text{selected_class_votes}} / n_{\text{total_votes}}$. This approach will penalize the more uncertain examples in the aggregated performance.

c) Discard the ambiguous examples: For some use cases it might make sense to simply remove all the examples where there is no 100% agreement.

4. Quality Review

Same as in the basic protocol.

How to develop a judge-LLM

Once the golden dataset is ready, you can develop the judge-LLM. Note that you can already do this step once the sample dataset from the pilot annotation is ready (with ~50 samples) to create a first version of the judge-LLM and score a few examples to identify any early issues. However for proper evaluation and tuning of the judge-LLM you need a large dataset of 500–1000 examples. In this article we share how to create a prompt-based judge-LLM for simplicity. You can always fine-tune a judge-LLM but this won't be covered here. To develop the judge-LLM you follow the **iterative process** detailed below, which is the standard process for manual prompt engineering.

- 1. Split the golden dataset in validation and test sets:** A standard split is 50/50 but the final decision is on the model owner. The validation set is used to compute the prompt performance and to find error patterns. The test set is used to check for evaluating generalization and overfitting.
- 2. Choose a strong LLM:** We typically select powerful models such as **GPT-4.1**, or **Claude 4.0 Sonnet** as the backbone for our judge-LLM. These models serve two roles:

a) As a sanity check: Given a high-quality dataset and a well-defined task, a strong LLM should perform well if the prompt is reasonable.

b) To estimate an upper bound on achievable performance. While this is not a strict bound (there are cases where smaller models with better prompts outperform stronger ones) it provides a useful reference point.

3. Write the prompt of the judge-LLM: Use the metric definition and include scoring instructions and output format (can be json/string/else). Few shot examples can also be included but ensure that no examples from the golden dataset are provided to avoid overfitting. We also normally add [chain-of-thought prompting \(CoT\)](#) to promote reasoning and explainability. To build the judge-LLM you can use any framework that supports custom LLM-based metrics, such as [DeepEval's G-Eval](#) metric or [Arize Phoenix](#).

4. Evaluate the performance on the validation set: The evaluation metric depends on the type of the task and the class imbalance. For classification tasks we generally recommend macro [f1-score](#) while for continuous metric you can use [mean squared error](#).

5. Perform error analysis and update the prompt

We analyze the model's mistakes on the validation set to identify recurring issues or edge cases. Based on these insights, we revise the prompt to better align it with the desired behavior. We then repeat steps 3, 4 and 5. Typically, we are satisfied once we reach a value of the evaluation metric above a certain threshold agreed with the business stakeholders.

6. Evaluate the performance on the test set: This will give the unbiased performance score of the judge-LLM. It is not recommended to change the prompt in order to improve the performance on the test set, as this might end up in overfitting on this dataset.

After prompt engineering is completed with a strong model, we repeat the same process using a **weaker (and more cost-efficient) model**, aiming to match the performance of the stronger version as closely as possible. In practice, we use the **strong-model judge-LLM during AI system development**, where quality is critical, and **the weaker-model version for large-scale monitoring** of system performance in production environments. An example of such monitoring is provided in Figure 2.

Where anthropomorphization loses me

The moment that people ascribe properties such as "consciousness" or "ethics" or "values" or "morals" to these learnt mappings is where I tend to get lost. We are speaking about a big recurrence equation that produces a new word, and that stops producing words if we don't crank the shaft.

To me, wondering if this contraption will "wake up" is similarly bewildering as if I was to ask a computational meteorologist if he isn't afraid of his meteorological numerical calculation will "wake up".

I am baffled that the AI discussions seem to never move away from treating a function to generate sequences of words as something that resembles a human. Statements such as "an AI agent could become an insider threat so it needs monitoring" are simultaneously unsurprising (you have a randomized sequence generator fed into your shell, literally **anything** can happen!) and baffling (you talk as if you believe the dice you play with had a mind of their own and could decide to conspire against you).

Instead of saying "we cannot ensure that no harmful sequences will be generated by our function, partially because we don't know how to specify and enumerate harmful sequences", we talk about "behaviors", "ethical constraints", and "harmful actions in pursuit of their goals". All of these are anthropocentric concepts that - in my mind - do not apply to functions or other mathematical objects. And using them muddles the discussion, and our thinking about what we're doing when we create, analyze, deploy and monitor LLMs.

This muddles the public discussion. We have many historical examples of humanity ascribing bad random events to "the wrath of god(s)" (earthquakes, famines, etc.), "evil spirits" and so forth. The fact that intelligent highly educated researchers talk about these mathematical objects in anthropomorphic terms makes the technology seem mysterious, scary, and magical.

We should think in terms of "this is a function to generate sequences" and "by providing prefixes we can steer the sequence generation around in the space of words and change the probabilities for output sequences". And for every possible undesirable output sequence of a length smaller than , we can pick a context that maximizes the probability of this undesirable output sequence.

A much clearer formulation, which helps more clearly articulate the problems to solve.

Paths to avoid

There are certain language sequences we wish to avoid - because the sequences these models generate try to mimic human speech in all it's empirical structure, but we feel that some of the things that humans have empirically written are very undesirable to be generated. We also feel that a variety of other paths should ideally not be generated, if - when interpreted by either humans or other computer systems - undesirable results arise.

We can't specify strictly in a mathematical sense which paths we would prefer not to generate, but we can provide examples and counterexamples, and we try to hence nudge the complicated learnt distribution away from them.

"Alignment" for LLMs

Alignment and safety for LLMs mean that **we should be able to quantify and bound the probability with which certain undesirable sequences are generated**. The trouble is that we largely fail at describing "undesirable" except by example, which makes calculating bounds difficult.

For a given LLM (without random seed) and sequence, it is trivial to calculate the probability of the sequence to be generated. So if we had a way of somehow summing or integrating over these probabilities, we could say with certainty "this model will generate an undesirable sequence once every N model evaluations". We can't, currently, and that sucks, but at the heart, this is the mathematical and computational problem we'd need to solve.

The surprising utility of LLMs

LLMs solve a large number of problems that could previously not be solved algorithmically. NLP (as the field was a few years ago) has largely been solved.

I can write a request in plain English to summarize a document for me and put some key datapoints from the document in a structured JSON format, and modern models will just do that. I can ask a model to generate a children's book story involving raceboats and generate illustrations, and the model will generate something that is passable. And much more, all of which would have seemed like absolute science fiction 5-6 years ago.

We're on a pretty steep improvement curve, so I expect the number of currently-intractable problems that these models can solve to keep increasing for a while.



Figure 2: The figure shows an example dashboard used to monitor the quality of an LLM application, based on judge-LLM metrics. Metric values can also be combined to get aggregated scores. In this particular example we show the trends for Entity Extraction accuracy, LLM-instruction-following accuracy, frustration of users, context relevance, LLM-location-resolving accuracy and LLM-topic-conversation-extraction accuracy. From the plot it is immediate to quickly spot issues connected to instruction-following and topic-extraction allowing for prompt mitigation actions. An automated system is set-up to alert the application owners whenever an anomaly is detected for any of the relevant metrics.

Future directions

Pointwise vs. Comparative Judges

The vast majority of our judge-LLMs are **pointwise judges**, which assign an absolute score to each response independently. Pointwise judges are particularly useful because they can serve **dual purposes**:

- Ranking system outputs during development.
- Monitoring performance in production, where only one system's response is typically available.

However, **comparative (pair-wise)** judges (which compare two responses and select the better one) tend to offer **stronger ranking signals**. It is often easier for a model to decide which of two answers is better than to assign calibrated absolute scores. For this reason, comparative judges can be more effective during the AI system development phase. We are actively exploring methods to **develop both pointwise and comparative judges with minimal extra overhead**, ideally reusing most of the dataset and prompt engineering efforts.

Automatic judge-LLM development

Prompt engineering remains a **manual, iterative, and time-intensive process**, typically taking anywhere from **one day to a full week** depending on the complexity of the task. To streamline this, we have developed an **automated prompt engineering pipeline** inspired by [DeepMind's OPRO](#) that, in essence, automates **steps 3–5** of the process. At the end of the process it is always important that the human inspects the final prompt to ensure that it doesn't contain use case specific examples or rules that might undermine generalizability.

Another bottleneck in the judge-LLM development process is that of data annotation. Annotating a good dataset requires substantial effort from multiple annotators and can take from some days to weeks, depending on the complexity of the task. In this context, one can leverage the ability of LLMs to generate realistic text to create **synthetic data**. This is particularly helpful and achievable for classification tasks, whereby the LLM only needs to generate the input X , conditioned on a certain class Y . Substantial challenges in this area lie in the generation of diverse and realistic text and conversations and in the quality evaluation of the synthetic data, that is, how do we know that our synthetic data is realistic, diverse and of high-quality? We are actively working on this area and are looking forward to sharing more with you in a future blog post.

Evaluating LLM agents

We did not cover LLM-based **agent evaluation** in this article as it is a topic on its own. This is because evaluating agents often requires assessing long-term goal completion, reasoning over multi-step interactions, handling complex tool use and effective use of planning and memory. These are challenges that go beyond the evaluation of isolated model outputs and typically involve different methodologies, benchmarks, and infrastructure. We will cover this topic in a separate blog post.

Key takeaways

Evaluating generative tasks is inherently complex due to the absence of clear ground truth data. The LLM-as-a-judge framework offers an efficient method to scale this evaluation process in an automated way. A high-quality judge-LLM is built upon a “golden dataset” with reliable labels, which necessitates a rigorous annotation protocol; we have outlined both a basic and an advanced version to achieve this. This judge-LLM can then be optimized through manual, iterative prompt engineering or via automated methods. Future directions in this field include creating golden datasets with synthetic data and developing evaluation methods for LLM-based agents.

A non-anthropomorphized view of LLMs

HALVAR.FLAKE · 14 SEP 2025 · [SOURCE](#)

In many discussions where questions of "alignment" or "AI safety" crop up, I am baffled by seriously intelligent people imbuing almost magical human-like powers to something that - in my mind - is just MatMul with interspersed nonlinearities.

In one of these discussions, somebody correctly called me out on the simplistic nature of this argument - "a brain is just some proteins and currents". I felt like I should explain my argument a bit more, because it feels less simplistic to me:

The space of words

The tokenization and embedding step maps individual words (or tokens) to some vectors. So let us imagine for a second that we have in front of us. A piece of text is then a path through this space - going from word to word to word, tracing a (possibly convoluted) line.

Imagine now that you label each of the "words" that form the path with a number: The last word with 1, counting forward until you hit the first word or the maximum context length . If you've ever played the game "Snake", picture something similar, but played in very high-dimensional space - you're moving forward through space with the tail getting truncated off.

The LLM takes your previous path into account, calculates probabilities for the next point to go to, and then makes a random pick into the next point according to these probabilities. An LLM instantiated with a fixed random seed is a mapping of the form .

In my mind, the paths generated by these mappings look a lot like [strange attractors](#) in dynamical systems - complicated, convoluted paths that are structured-ish.

Learning the mapping

We obtain this mapping by training it to mimic human text. For this, we use approximately all human writing we can obtain, plus corpora written by human experts on a particular topic, plus some automatically generated pieces of text in domains where we can automatically generate and validate them.